Design to Cost
User guide

Part, raw material, tool and mould cost from the 3D model, by country, with every value traceable.

Version V0 · 13 November 2026 · covers roadmap stages 1-56

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1. About this guide

Design to Cost (DTC) works out the should-cost of a part from its 3D model: raw material (RM), the tool or mould and its CAPEX, the part cost per piece, and the same in other countries with the landed cost in India. Every number can be clicked to see how it was made. Nothing is hard-coded: rates, rules and formulas are master data that you can change.

This guide explains every page and the monthly routine. For a step-by-step worked example (L bracket, correction, Rev B, plastic cover with its mould) see the Work instruction on the Help page.

Flow of a part	Where
1. Import the 3D model (STEP), choose main and sub material	Import
2. RM cost: gross weight × approved material rate − scrap credit	Parts → part panel → RM cost
3. Tool / mould cost, tool design, drawings, simulation	Parts → part panel → Tool cost, Tool design
4. Part cost per piece: RM + process + rejection + overhead + profit + packing + tool amortisation + logistics	Parts → part panel → Part cost
5. Project totals, BOM cost, CAPEX, cost changes	Dashboard

All amounts are shown INR first. Values marked EXAMPLE (orange) are placeholders shipped with the tool: replace them with your own (Master data → Replace example values).

2. Install, start, update and back up (Windows)

The app runs on your PC at http://127.0.0.1:8100. The kit files live in the base folder, e.g. D:\Project\DTC Tool and mould:

Double-click	What it does
DTC_Start.bat	Starts the app (a backup first if the last one is older than a day). Keep the black window open; close it to stop.
DTC_Update.bat	Backup → new version from GitHub (1) or a ZIP (2) → libraries → your master data merged with the new rows → report.
DTC_Backup.bat	Backup now into DTC_Backups (and the copy folder of dtc_settings.ini).
DTC_Restore.bat	Puts a backup back: everything, only projects + CAD files, or only master data.
DTC_Rollback.bat	Back to the version before the last update, with the data of that moment.
DTC_ResetPassword.bat	New password for a login ID (e.g. a forgotten admin password); only at this PC.
DTC_Country_Rates.bat	Monthly rate update + country rate book (one Excel per country) into DTC_Data\country_rates.
DTC_Install_FEA.bat	Installs Gmsh for the strength check (FEA); CalculiX ccx.exe goes into DTC_Data\tools\calculix.

Where your data is

- DTC_Data\master_data: rates, machines, formulas (costing_db.sqlite, master_data.xlsx).
- DTC_Data\data: projects, parts and every cost version (projects.sqlite); DTC_Data\storage: CAD files.
- An update never overwrites your data: new rows are added, rows you changed stay yours, rate versions are never changed.
- Set backup_copy_to in dtc_settings.ini to a OneDrive or network folder so every backup also leaves the PC.

Every update

1. Close the app (its black window) and any open DTC Excel file.
2. Double-click DTC_Update.bat, choose 1 (download from GitHub), wait for UPDATE DONE.
3. Double-click DTC_Start.bat. Check one known part against its last cost.

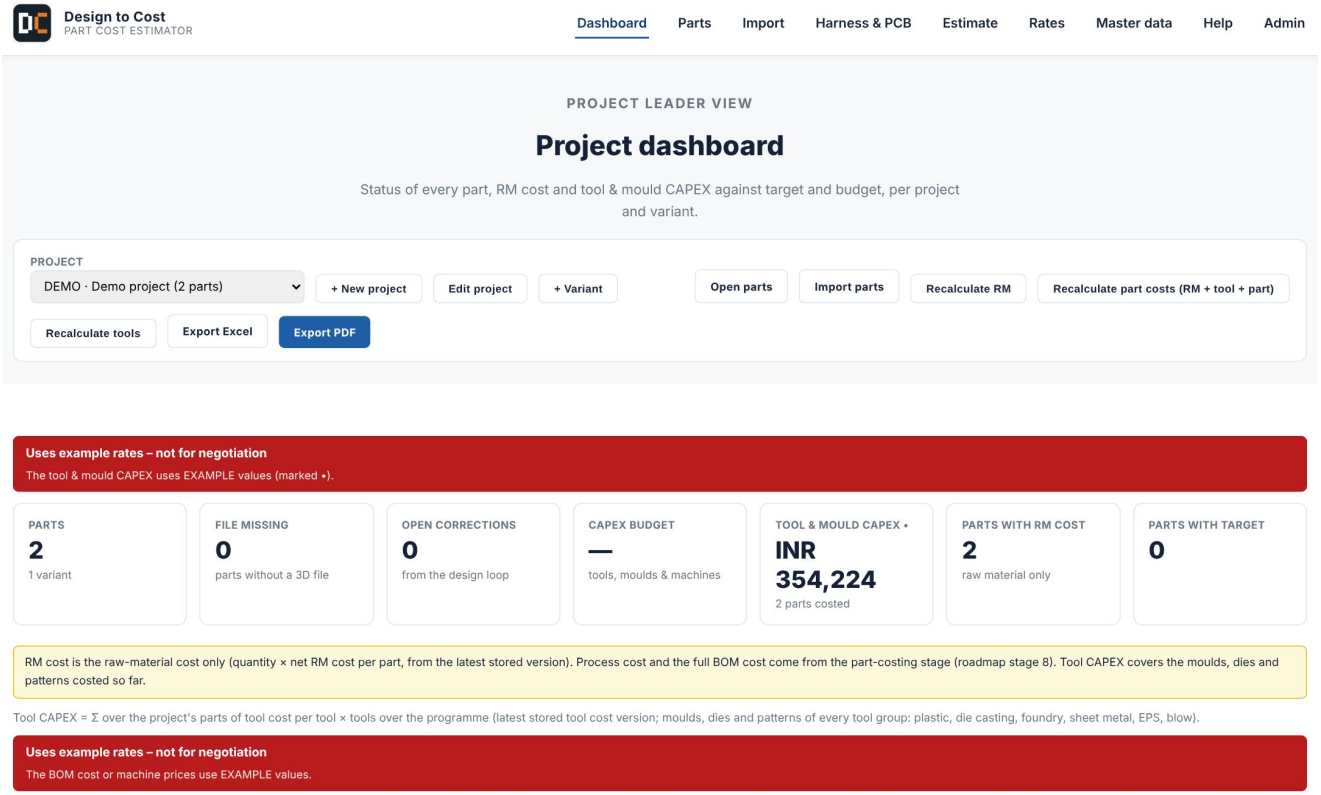
3. Logins, roles and tool groups

Everybody logs in with an own ID; every change, approval and backup is recorded under it. The first start asks for the admin ID. Roles: viewer < engineer < approver < head < admin.

Role	May
viewer	read everything
engineer	projects, parts, costs, imports, master data, rate drafts
approver	+ approve rates, see the activity log, delete parts
head	sees and changes the data of every tool group and user (no user management)
admin	+ login IDs, permissions, backups, app updates, AI assistant, monthly rates

- Admin → Users, roles & permissions: per-user Allow / Deny on top of the role, and tool groups (Plastic moulding, Die casting, Foundry, Sheet metal, EPS, Blow moulding, Machining). A user with groups sees only those parts.
- The permission Open traces (admin by default) lets a user click a value to see its full calculation.

4. Dashboard: the project



Dashboard: project tiles, variants, BOM cost and the recalculation buttons.

- + New project (code, name, customer, CAPEX budget, SOP date) and + Variant (target BOM cost, annual volume).
- Tiles: parts, files missing, open corrections, BOM cost against target, tool CAPEX against budget.
- Project leader view: BOM cost per variant, CAPEX plan with milestones, parts by status / process / sourcing.
- Export PDF (project summary) and Excel (part register); totals per tool group for limited users.

Recalculate

Button	What it does
Recalculate part costs (RM + tool + part)	One click: RM cost, tool cost and part cost of every part with the latest approved rates. Only parts whose cost changed get a new version.
Recalculate RM / Recalculate tools	Only that step, for every part.

Stored costs never change by themselves: a new price only reaches a part when you recalculate. Old versions are kept.

5. Cost changes report



PART	NAME	PART COST ON 2026-10-14	PART COST NOW	CHANGE (₹/PART)	CHANGE (%)	STATUS	WHY (COST LINES)
DEMO-0001	plastic_cover	₹16.36 v1 · 2026-10-04 15:33	₹19.76 v2 · 2026-11-13 15:33	+₹3.40	+20.8 % ⚑	Changed	▼ 4 line(s) moved
DEMO-0002	sheet metal bracket	₹10.04	₹10.04	₹0.00	0.0 %	Unchanged	▶ 0 line(s) moved

COST LINE	THEN	NOW	CHANGE
Net RM cost	**₹11.92**	**₹14.90**	**+₹2.98**
Process cost	₹1.67	₹1.67	₹0.00
Rejection cost	**₹0.28**	**₹0.34**	**+₹0.06**
Overhead	₹0.08	₹0.08	₹0.00
Profit	**₹1.39**	**₹1.70**	**+₹0.30**
Packing	**₹0.23**	**₹0.28**	**+₹0.05**
Tool amortisation	₹0.33	₹0.34	+₹0.00
Logistics	₹0.45	₹0.45	₹0.00

4 line(s) at ₹0 then and now not shown.

recalculated with the latest approved rates · Guide · rates of 2026-11

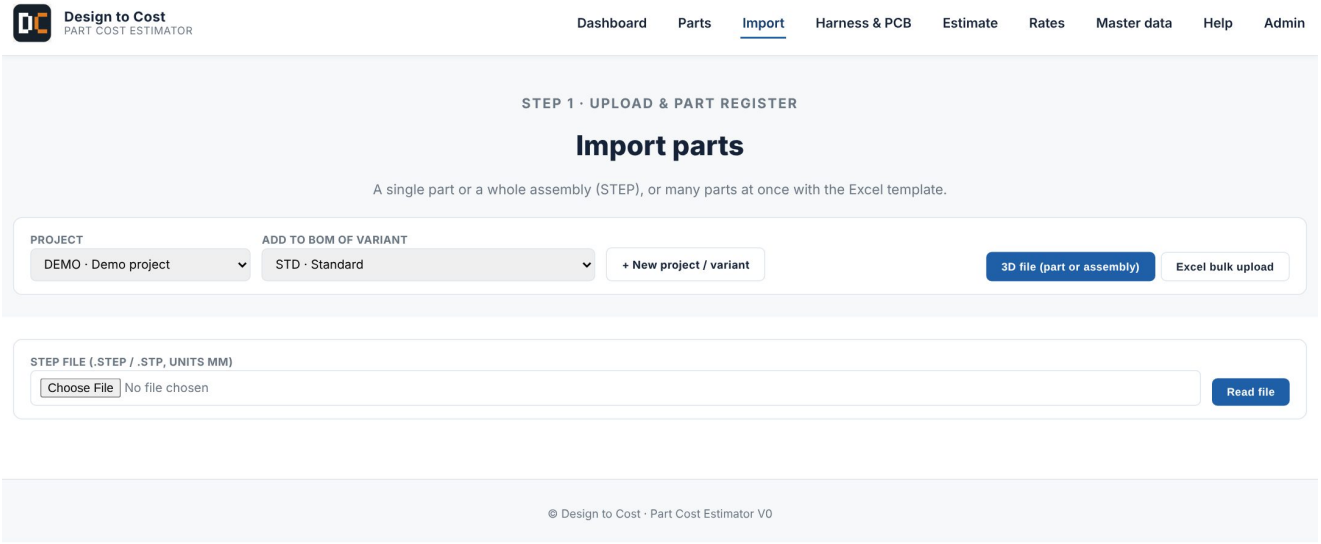
⚑ = change of ±5% or more (Master data → Parameters → all / cost_change_flag_pct). Each cost line is read from that version's own trace.

Cost changes: part cost now against the cost valid on a date, with the cost lines that moved.

- Pick a date (default: 30 days back, Parameters → cost_change_lookback_days) and press Show changes.
- Per part: cost on that date (version, date), cost now, change in ₹/part and %, status (changed, new since the date, unchanged, no part cost yet).
- ⚑ marks a change of ±5 % or more (Parameters → cost_change_flag_pct); flagged parts come first.
- Why: the cost lines then / now / change (net RM, process, rejection, overhead, profit, packing, tool amortisation, logistics). They always add up to the part cost.
- Excel and PDF download the same report for the chosen date.

Typical use: after a price change in the Price master, click Recalculate part costs, then Show changes, and send the PDF.

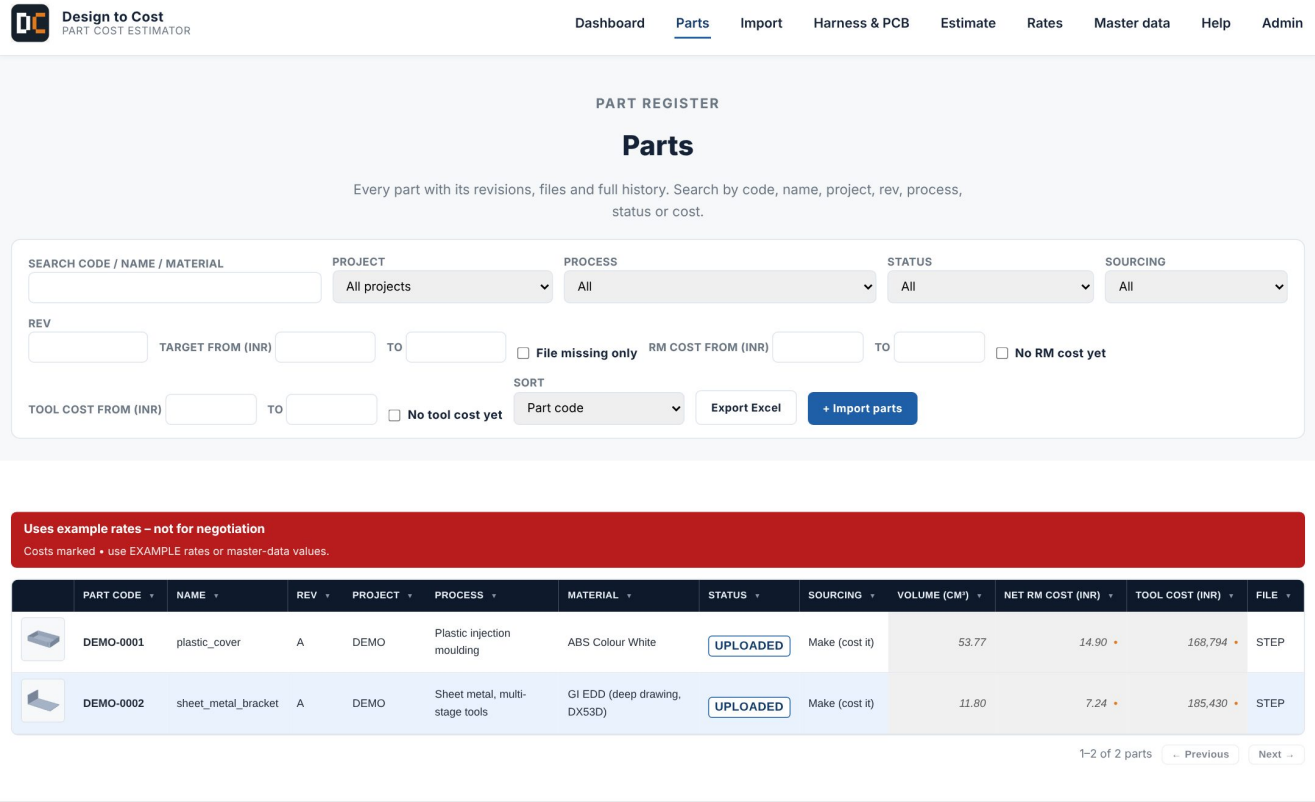
6. Import: bring in parts



Import: a STEP file or an assembly; the tree shows every part with its thumbnail and quantity.

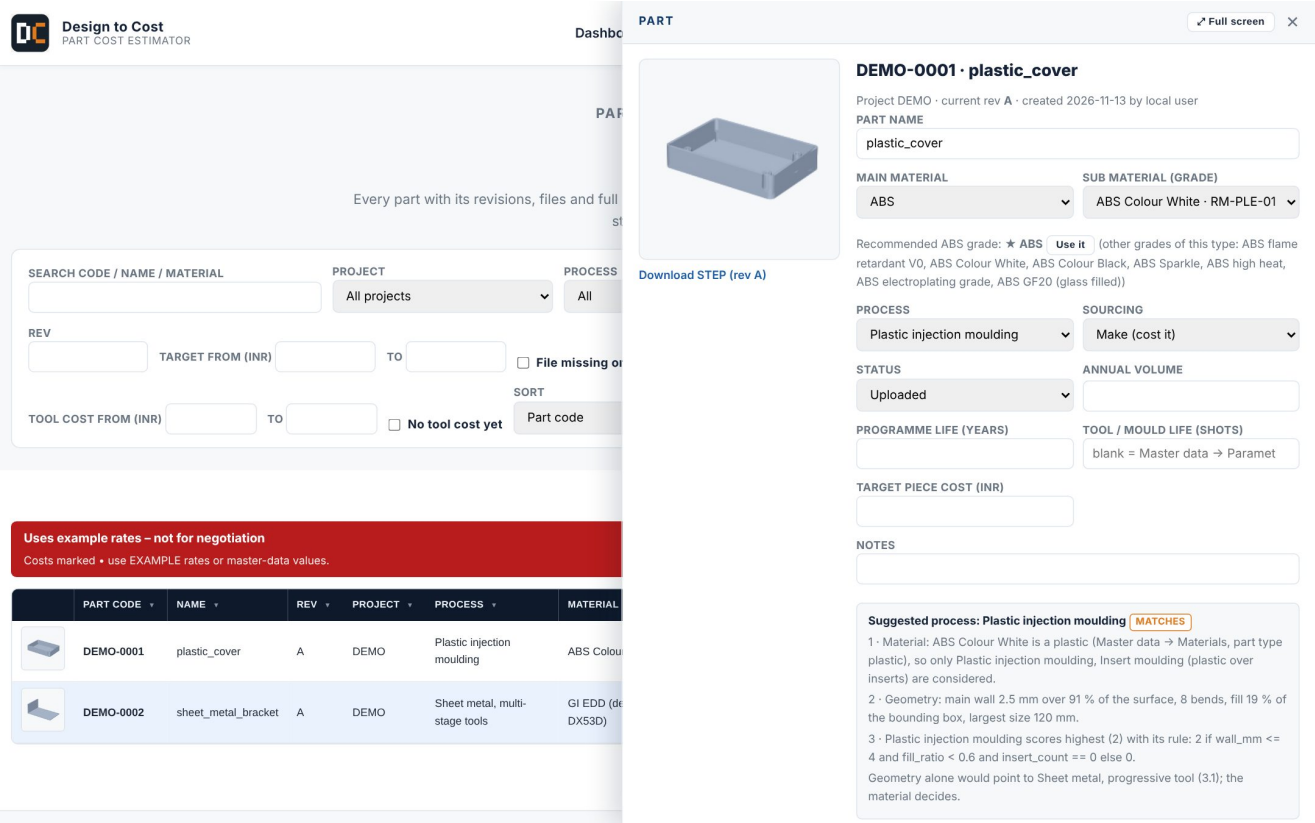
- 3D file (STEP): a part or an assembly. An assembly is split into its parts; identical parts become one part with a quantity; look-alikes are flagged for you to decide.
- For each part: code (blank = generated), name, main material → sub material (grade), process and sourcing (Make / Bought-out / Existing / Not required).
- The process is suggested with its reason (why?): material family first, then the geometry, then moulded-in inserts.
- Excel bulk upload: download the template, one row per part, upload with the STEP files or a ZIP. Nothing is saved if any row is wrong.
- Scanned drawings and DWG are read on the Estimate page.

7. Parts: the register and the part panel



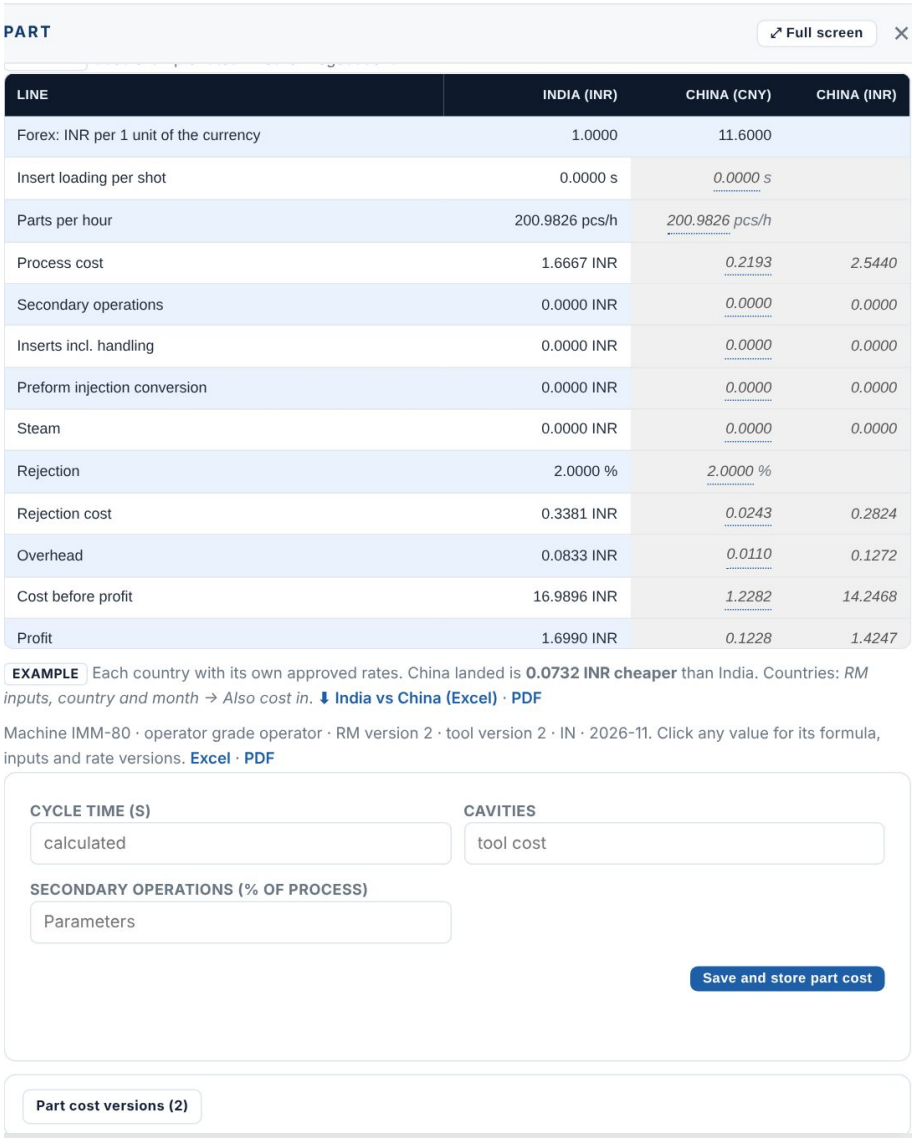
Parts: search, filters (▾ in each column header), costs per part.

Click a part to open its panel. The panel holds everything about the part, top to bottom:



Part panel: details with main material → sub material (with its code), process, volume, country.

Block	What you see and do
Details	Name, main / sub material, process, sourcing, volume, programme life, tool life, producing country, Also cost in. Every change goes into the history.
Geometry	Volume, net weight, projected area, size, wall, bends (click for the trace). Part drawing PDF / DXF.
Design loop	Status flow (Uploaded → Concept tool → Simulation → … → Tool design frozen), comments, revisions, compare two revisions.
RM cost	Gross weight, yield, rate, scrap credit, net RM cost per country; type over any input; Excel / PDF cost sheet.
Tool cost	Cavities, mould base or die set, BOM, process hours × MHR, design, trial, overhead, profit; per die stage for sheet metal; progressive vs stage dies.
Tool design	3D design, drawing set (one page per drawing), 3D viewer, STEP / STL, release ZIP, toolmaker review, strength check (FEA).
Simulation	Wall thickness, draft, undercuts, flow simulation (fill, pressure, cooling, weld lines, air traps), report PDF and video.
Part cost	RM + process + rejection + overhead + profit + packing + tool amortisation + logistics, for every country; versions; Excel / PDF.
Country comparison	The part and its tool made in each country, ex-works and landed cost in India (freight, insurance, duty, SWS, IGST).
Saved files	Every download is kept with a revision; an old one is served again until something it depends on changes.

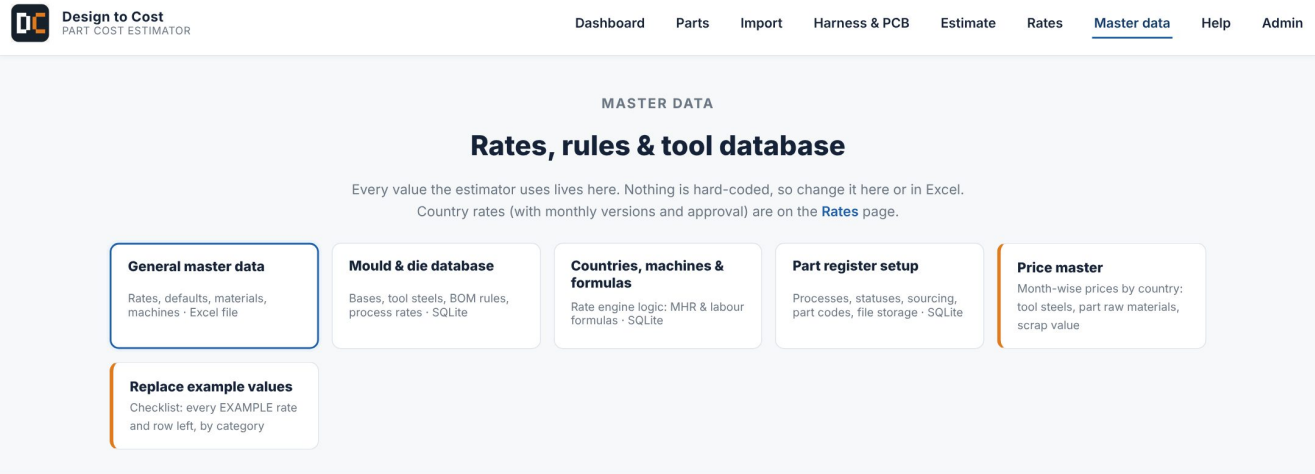


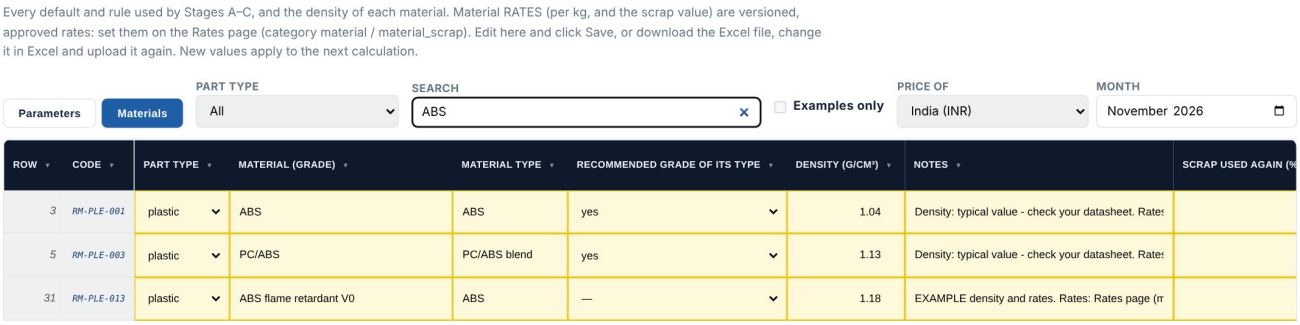
Part cost of a part: each line can be clicked for its trace, down to the electricity rate in the MHR.

Correct any value

- Click a value → the traceability panel shows the formula, every input and its source (rate version, month, approval).
- Change an input there (typed value) and save: a new version is made; the old one stays.
- A blocking toolmaker-review point stops Tool design frozen until it is solved or waived.

8. Master data





Master data → General master data → Materials: unique code, main material (material type), sub material (grade) and the live price.

Every value the estimator uses lives here. Change it on the page and click Save, or download the Excel file, change it, upload it.

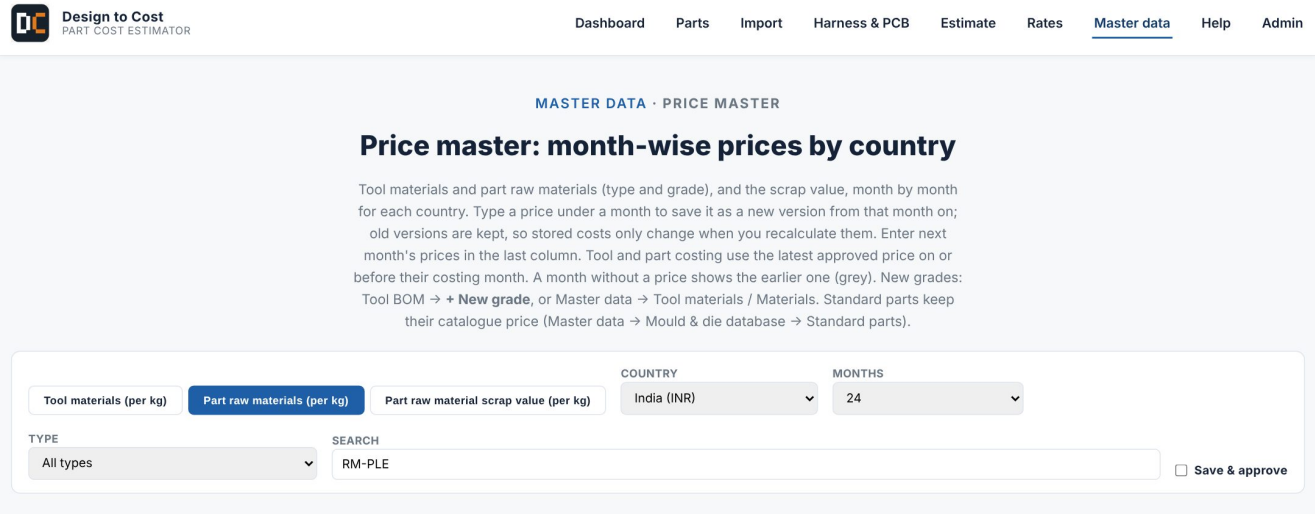
Card	What is in it
General master data	Parameters of each part type, Materials (main material, grade, density, scrap reuse %).
Mould & die database	Mould bases, tool steels, BOM rules, design rules, sheet-metal and strip rules, flow materials, process windows, mould systems, review checklist.
Countries, machines & formulas	Countries and currencies, machines, labour grades, every MHR / labour / RM / tool / part-cost formula and its inputs.
Part register setup	Processes and their rules, part statuses, sourcing types, insert materials and shapes, part-code pattern, register settings.
Price master	Month-wise prices per country: tool steels, part raw materials, scrap value.
Replace example values	Checklist of every EXAMPLE rate and row still left, by category.

Unique codes and live prices

- Every row has a code that never changes or is reused: part materials by family (RM-STL carbon / alloy steel, RM-SHT sheet, RM-SS stainless, RM-ALC / RM-ALW aluminium cast / wrought, RM-ZN zinc, RM-CU copper / brass, RM-CI cast iron, RM-PLC / RM-PLE plastics), TS tool steels, MC machines, PR / PU process rates, others by table initials (e.g. DR design rules).
- Materials, tool steels, machines and process rates show a live Price column for the country and month you pick above the table. Prices are never copied: change one in the Price master and every page shows it at once. edit opens the Price master on that code.

- Main material → sub material: the material type is the main material (e.g. ABS) and the grade the sub material (e.g. ABS Colour White · RM-PLE-015). New grades start with the price of their main grade, marked EXAMPLE.

9. Price master

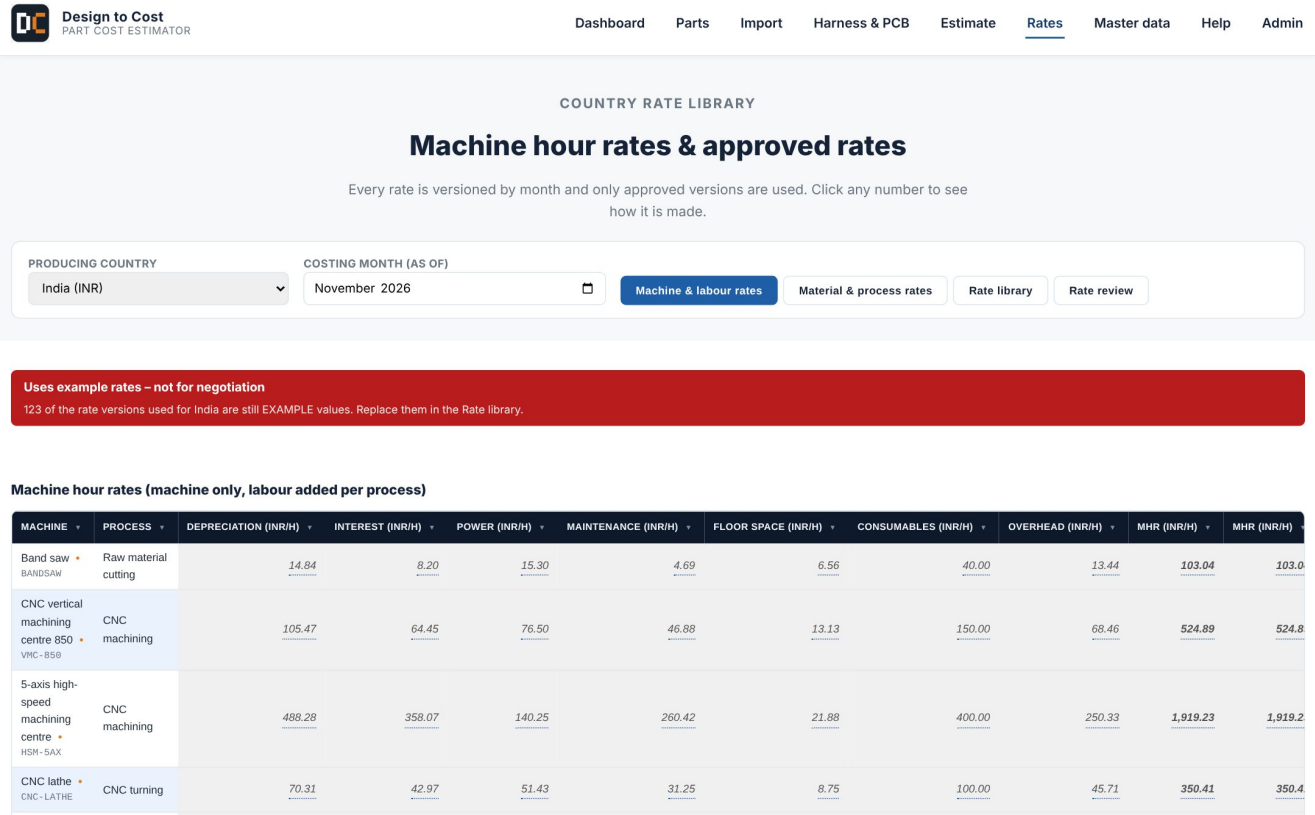


	2025-08	2025-09	2025-10	2025-11	2025-12	2026-01	2026-02	2026-03	2026-04	2026-05	2026-06	2026-07	2026-08	2026-09	2026-10	2026-11	2026-12 NEXT
RM-PLE-001 ABS			★ ABS			157	160	167	177	202	194	185	183	180	180	180	180
RM-PLE-002 Polycarbonate (PC)			★ PC (polycarbonate)			227	231	241	255	292	280	268	264	260	260	260	260
RM-PLE-003 PC/ABS blend			★ PC/ABS			200	204	213	226	258	248	237	233	230	230	230	230
RM-PLE-004 Polyamide (PA)			★ PA6			218	222	232	245	281	269	257	254	250	250	250	250

Price master: part raw materials of India, 24 months; Code · main material, sub material.

- Pick the group (tool materials, part raw materials, scrap value), the country and the months.
- A cell shows the approved price costing uses that month: yellow = a version starts there, grey = carried over.
- Type a price under a month: a new version from that month on (a draft; approvers can tick Save & approve).
- Excel: download the grid, change prices, upload: only changed cells become new versions.
- Months before your first price are filled from the market (24-month history, see Monthly rates); they say INDICATIVE and EXAMPLE. A month you enter yourself is never overwritten.

10. Rates



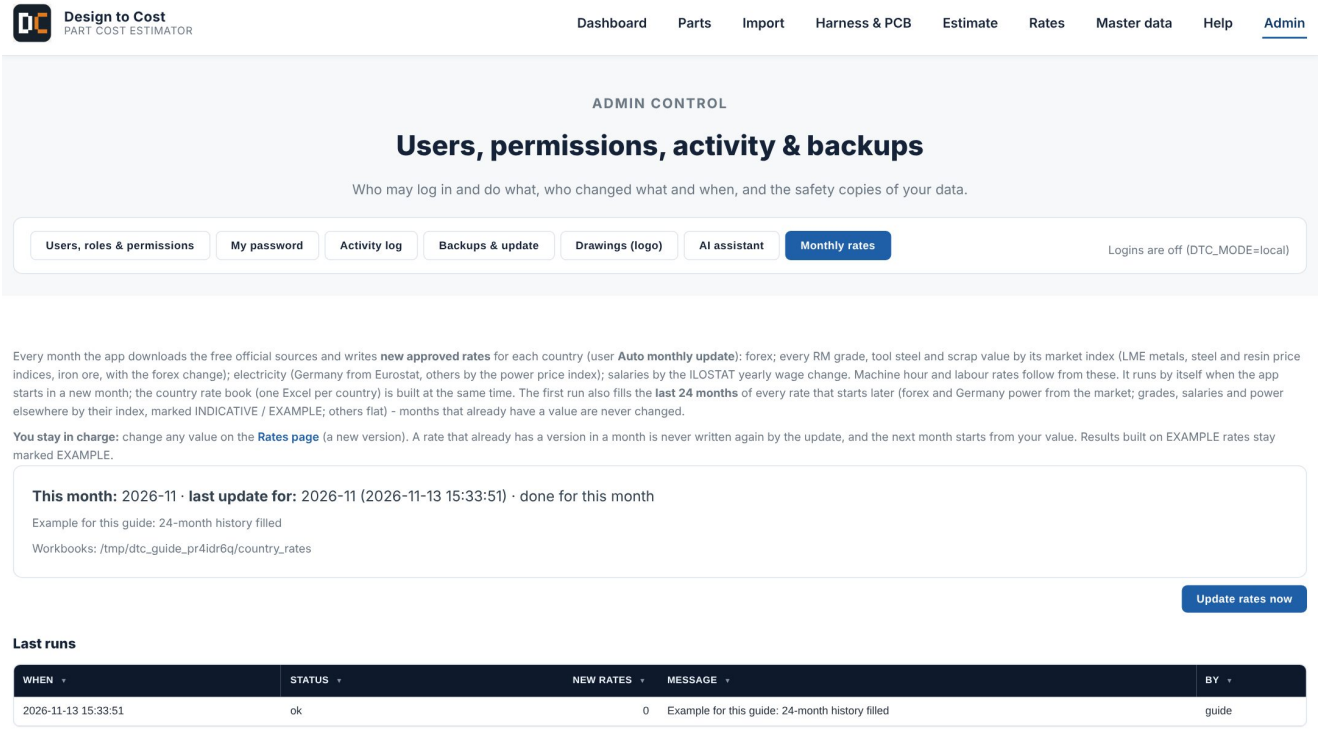
Rates: machine hour rates and labour rates of a country and month, with their traces.

- Pick the producing country and costing month.
- Machine & labour rates: MHR of every machine and labour rate per skill level, in local currency and INR. Click a number for its trace.
- Rate library: every approved rate; new version (draft or Save & approve), Excel export with history, Excel import as drafts.
- Rate review: drafts next to the approved value, % change (highlighted above 5 %), approve / review / reject one by one or in bulk.

Formula	
MHR	depreciation + interest + power + maintenance + floor space + consumables + overhead, per hour
Labour rate	monthly salary × 12 × (1 + employer %) ÷ (days × hours) × (1 + supervision %) ÷ productive %

Costing uses only the latest APPROVED version up to the costing month. Approved versions are never changed or deleted.

11. Monthly rates: automatic every month



Admin → Monthly rates: last run, Update rates now, and the log.

When the app starts in a new month it downloads the free official sources and writes new APPROVED rates for every country (user Auto monthly update). Admin → Monthly rates → Update rates now, or DTC_Country_Rates.bat, does the same at any time.

Rate	How it is updated	Source
Forex	market monthly average	FRED (Federal Reserve), World Bank for VND
RM grades, tool steels, scrap values	last rate × the grade's market index × forex change	LME / IMF metals, US steel and resin price indices
Electricity	Germany: Eurostat; others: power price index	Eurostat nrg_pc_205, US PPI
Salaries	× ILO yearly wage change when a new year is published	ILOSTAT
Machine hour and labour rates	follow by formula	—

- You stay in charge: change any value on the Rates page or the Price master. A month that already has a value is never written again, and the next month starts from your value.
- 24-month history: every run also fills the months still empty in the last 24 (forex and Germany power from the market; grades, salaries and other rates INDICATIVE by their index, others flat). They are marked EXAMPLE.
- Results built on EXAMPLE rates stay EXAMPLE until you replace the base value.

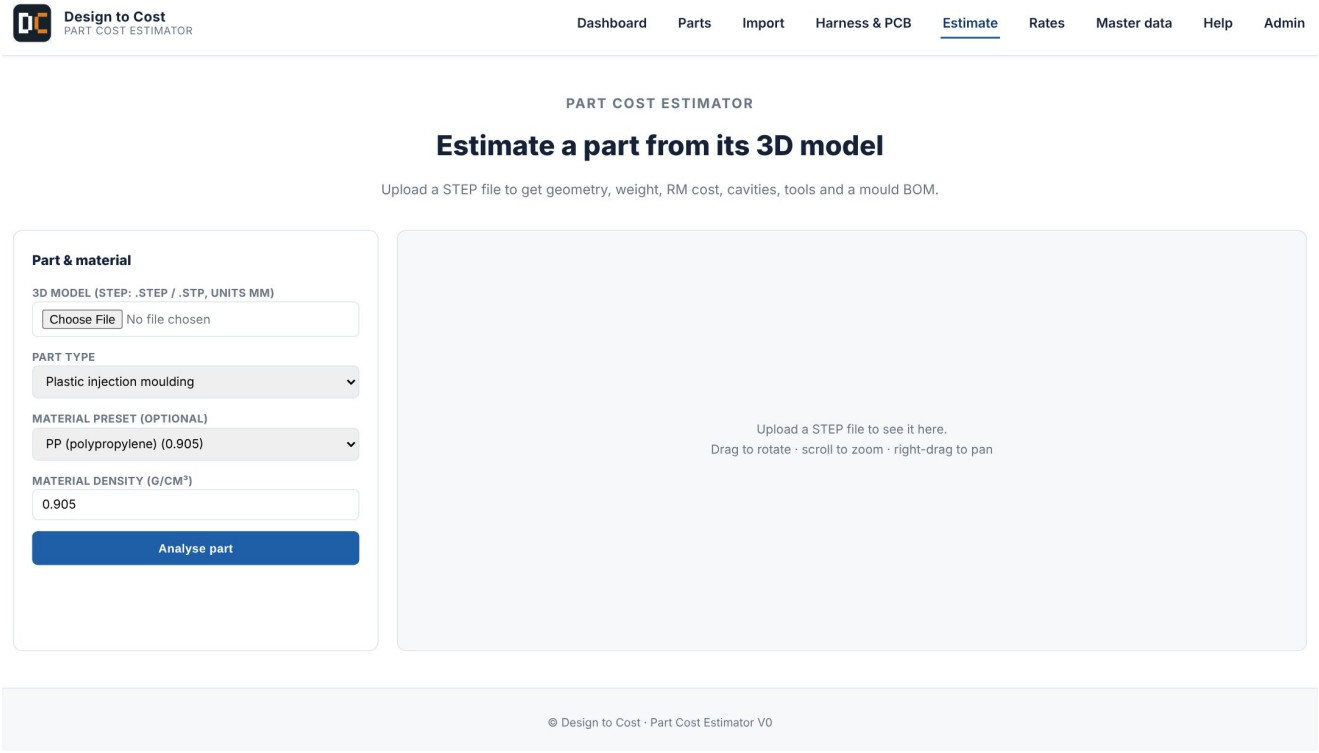
12. Country rate book (Excel per country)

DTC_Country_Rates.bat (or the monthly update) writes one workbook per country into DTC_Data\country_rates, 24 months:

Sheet	What is in it
Summary	Latest value, change over 12 months and since the first month, data gaps, EXAMPLE count.
RM_Price	A: world metal prices in USD, INR/kg and local currency/kg; B: every grade of the tool by family; B2 / B3: indicative 24-month grade trend; C: more grades for your quotes.
Labour	ILO hourly wage by skill level and manufacturing monthly wage; the tool's labour rates.
Process_Cost	Machine hour rate of every machine, process rates, cost drivers.
Electricity	The tool's power rate, Eurostat (EU), official tariff publisher, entry rows.
Transport	Freight rates, Brent, diesel, freight indices, entry rows for your quotes.
Forex	INR per USD and per local currency; check against the tool's forex.
Sources	Publisher and link of every series, refresh log.

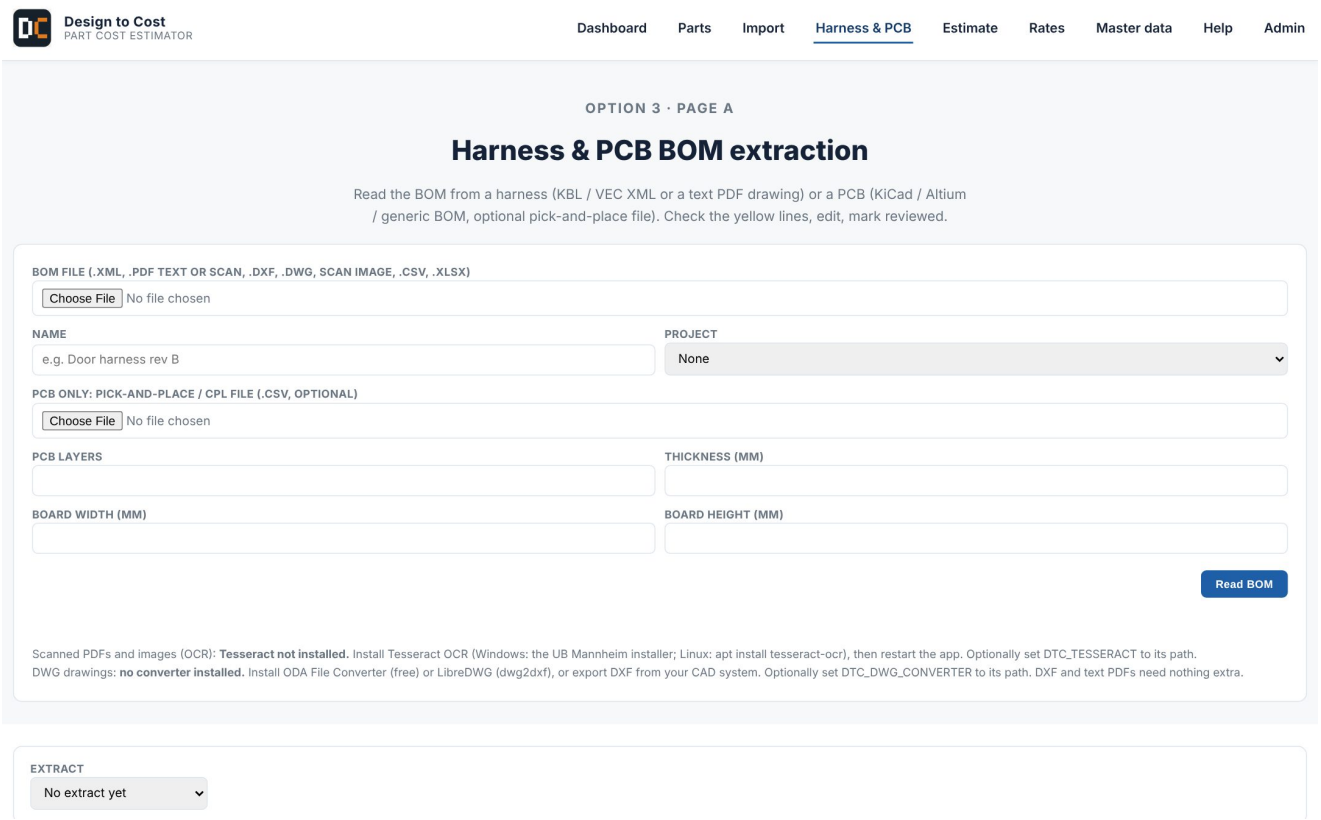
- Green = downloaded market data, blue = approved rate of the tool, orange = EXAMPLE, yellow = your entry, lilac = indicative.
- ILOSTAT refuses automatic downloads: download its two CSV files in the browser and run DTC_Country_Rates.bat --ilo-file "C:\Users\<you>\Downloads" once a year.
- Close the workbooks in Excel before a refresh; an open one is skipped (NOT written, close it and run again).

13. Estimate, Harness & PCB



Estimate (start page): a quick RM, tool and part cost from a 3D file or a few inputs, without registering the part.

- Estimate: upload a STEP (or a scanned drawing / DWG), choose material, process and country: Stage A (geometry, weight), Stage B (RM cost), Stage C (cavities, tools). Open in Estimate from a part loads its inputs; Save to part stores them.
- Harness & PCB: reads a harness or PCB BOM (Excel), groups wires, terminals, connectors and components, and costs them with the rate library (consumables, conversion, bare board).



Harness & PCB: read a harness or PCB BOM and cost it with the rate library.

14. Admin

Tab	What it does
Users, roles & permissions	Login IDs, roles, per-user Allow / Deny, tool groups, block / unblock, new password (admin only).
My password	Change your own password.
Activity log	Who changed what and when; filters; Excel export.
Backups & update	Installed version, Backup now, download a backup, update reports, Update app.
Drawings (logo)	Logo, company and title-block data on every drawing, Excel and PDF.
AI assistant	Ask AI on every page (read only, with the user's own permissions); needs an Anthropic API key.
Monthly rates	Last automatic update, Update rates now, the log of each run.

15. Monthly routine (checklist)

1. Start the app in the new month: the rates of the month and any missing history are written by themselves (Admin → Monthly rates shows the result).
2. Enter real supplier prices in the Price master where you have them (they replace the indicative / EXAMPLE values).
3. Rates → Rate review: approve open drafts.
4. Dashboard → Recalculate part costs (RM + tool + part) for each active project.
5. Dashboard → Cost changes → Show changes; download the PDF for your team.
6. Optional: DTC_Country_Rates.bat for the country workbooks; DTC_Backup.bat before big changes.

16. Troubleshooting

What you see	What to do
Permission denied …Country_Rates_….xlsx	The workbook is open in Excel: close it and run again.
NO market index data downloaded yet	The PC could not reach the sources (FRED, Eurostat, World Bank): check internet / proxy, then Update rates now.
FAILED ILOSTAT … 403	Normal: ILOSTAT refuses automatic downloads; your saved wage data is used. New year: --ilo-file.
No approved rate '…' for …	Add or approve that rate on the Rates page (or the Price master) for the country and month.
Recalculate says unchanged	No input changed since the last version; check the costing month and the country of the part.
'No module named backend'	Use the .bat files in the base folder, never python -m … from another folder.
Forgot the password	DTC_ResetPassword.bat at the PC that runs the app.
Something wrong after an update	DTC_Rollback.bat puts the previous version and its data back.

17. Words used

Word	Meaning
RM	raw material; net RM cost = RM cost − scrap credit
MHR	machine hour rate (₹ per hour of a machine)
CAPEX	tool / mould and machine investment
Approved / draft	only approved rates are used; new prices start as drafts unless approved at once
EXAMPLE	placeholder value shipped with the tool, to be replaced with yours
INDICATIVE	a past month worked out from a market index, not a quote
Version	every stored cost is a numbered version; old versions never change
Trace	the full calculation behind a value: formula, inputs, sources